Quizrr Advanced Full Test 10 (Paper 1) – with Video Solutions - 2025

Total Time: 180 minutes | Total Marks: 195 | Total Questions: 54

GENERAL INSTRUCTIONS

1. The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **54** questions.
2. Each subject is divided into three sections:
   - **Section 1:** Single Correct Type Questions (4 options, only one correct)
   - **Section 2:** Multiple Correct Type Questions (4 options, one or more correct)
   - **Section 3:** Numerical Value Type Questions (answer to be entered as numerical value)
3. Marking Scheme:
   - **Single Correct:** +3 for correct, -1 for wrong, 0 if unattempted.
   - **Multiple Correct:** +4 for all correct options, partial marking +1 otherwise, -2 for wrong options.
   - **Numerical Value:** +3 for correct, 0 for wrong/unattempted, no negative marking.

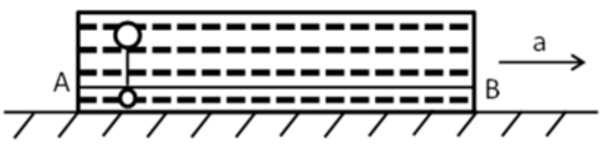
By: C I P H Ξ R

Syllabus: Full Syllabus

SECTION 1 - Physics [SINGLE CORRECT TYPE]

Q.1

A tubular container contains a homogenous non viscous liquid. Wire AB is attached inside the tubular container. A uniform ball of density ρ is attached to wire AB by means of a cord of negligible mass and a ring. Ring is able to move without friction.



Initially system was at rest and the ball is at left end of the wire. At time $t = 0$ the tubular container is accelerated horizontally with uniform acceleration 'a' towards right. The distance travelled by container in the time ball reaches the end B of the wire is N times length of wire AB, then density of liquid is :-

(A) $\frac{1+N}{N}\rho$

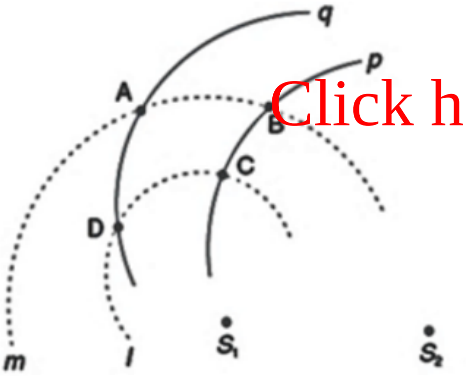
(B) $\left(\frac{1+N}{N}\right)\frac{g}{a}\rho$

(C) $\frac{N}{N+1}\rho$

(D) $\left(\frac{N}{N+1}\right)\frac{g}{a}\rho$

Q.2

In the figure shown, S_1 and S_2 are two identical point sources of sound which are coherent 180° out of phase. Taking S_1 as centre, two circular arcs ℓ and m of radii 1 m and 2 m are drawn. Taking S_2 as centres, two circular arcs p and q are drawn having radii 2 m and 4 m respectively. Out of the four intersection points A, B, C and D which point will record maximum intensity and which will record the least intensity of sound respectively? It is given that wavelength of wave produced by each source is 4.0 m.



(A) Maximum at C, least at A

(B) Maximum at C, least at B

(C) Maximum at A, least at B

(D) Maximum at D, least at B

Q.3

A parallel beam of uniform monochromatic light of wavelength 546 nm has an intensity of 200 $\mathrm{W/m^2}$. The number of photons in 1 $\mathrm{mm^3}$ of this radiation is:

(A) 183

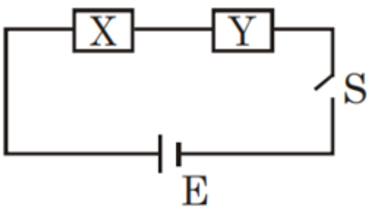
(B) 1831

(C) 18311

(D) 183117

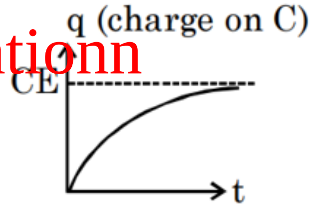
Q.4

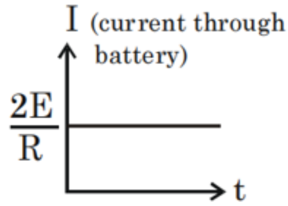
Following is a circuit containing an ideal battery of emf E in which the elements X & Y are unknown & may be a capacitor C, a resistor R, a battery of emf E, or may be their combination as mentioned in each case. The connecting wires are perfectly conducting.

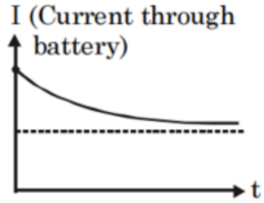


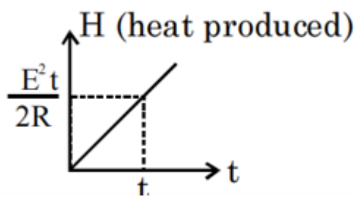
The switch S is initially open & it is closed at $t = 0$.

Column-1		Column-2		Column-3
(I)	Element X is R Element Y is C	(i)	Entire energy (non-zero) supplied by batteries is dissipated as heat	(P)
(II)	Both elements X & Y are R each	(ii)	Total charge drawn from the battery(s) after long time is $\frac{CE}{2}$	(Q)
(III)	Element X is R, element Y is a parallel combination of R and C	(iii)	Total heat produced in circuit is less than the energy supplied by the battery(s)	(R)
(IV)	Element X is R element Y is battery E joined with same polarity	(iv)	I = 0 just after S is closed as well as after long time	(S)

(P) 

(Q) 

(R) 

(S) 

Of the given 4 situations, which one has maximum value of current just after closing the switch?

(A) I, iii, P

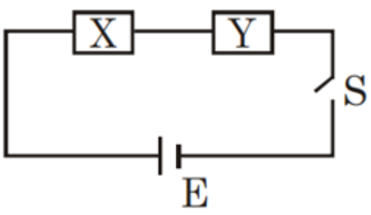
(B) IV, i, Q

(C) II, iii, P

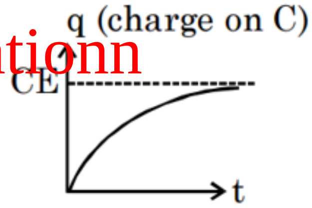
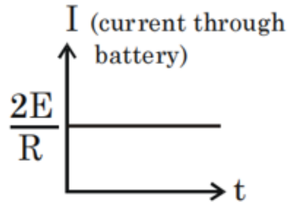
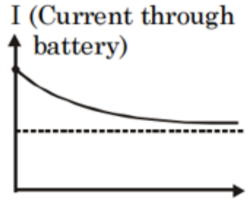
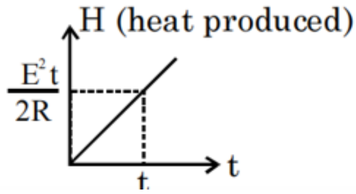
(D) III, i, S

Q.5

Following is a circuit containing an ideal battery of emf E in which the elements X & Y are unknown & may be a capacitor C, a resistor R, a battery of emf E, or may be their combination as mentioned in each case. The connecting wires are perfectly conducting.



The switch S is initially open & it is closed at $t = 0$.

Column-1	Column-2	Column-3
(I) Element X is R Element Y is C	(i) Entire energy (non-zero) supplied by batteries is dissipated as heat	(P) 
(II) Both elements X & Y are R each	(ii) Total charge drawn from the battery(s) after long time is $\frac{CE}{2}$	(Q) 
(III) Element X is R, element Y is a parallel combination of R and C	(iii) Total heat produced in circuit is less than the energy supplied by the battery(s)	(R) 
(IV) Element X is R element Y is battery E joined with same polarity	(iv) $I = 0$ just after S is closed as well as after long time	(S) 

In which case the charge drawn from the battery (s) during $t = 0$ to $t = RC$ is the least?

(A) II, i, S

(B) I, iii, P

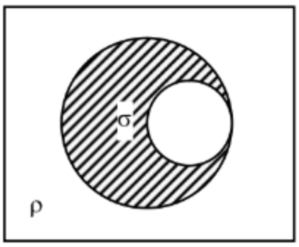
(C) III, ii, Q

(D) IV, i, Q

Q.6

Passage:

A uniform solid sphere of density σ and radius R has a spherical cavity of radius $\frac{R}{2}$ between the centre and surface of the sphere. It is immersed in a tank containing liquid of density ρ with the liquid level $\frac{R}{4}$ above the sphere top. The sphere is released from rest at the position shown with centre of cavity and the centre of the sphere at the same horizontal level. It is observed that centre of mass of sphere stays at rest.



Question:

Relation between ρ and σ is :

(A) $\frac{\sigma}{\rho} = \frac{8}{7}$

(B) $\frac{\sigma}{\rho} = \frac{7}{8}$

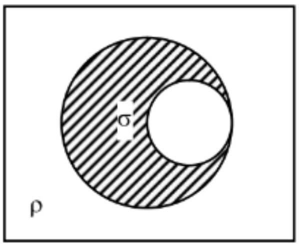
(C) $\frac{\sigma}{\rho} = 1$

(D) $\frac{\sigma}{\rho} = \frac{16}{15}$

Q.7

Passage:

A uniform solid sphere of density σ and radius R has a spherical cavity of radius $\frac{R}{2}$ between the centre and surface of the sphere. It is immersed in a tank containing liquid of density ρ with the liquid level $\frac{R}{4}$ above the sphere top. The sphere is released from rest at the position shown with centre of cavity and the centre of the sphere at the same horizontal level. It is observed that centre of mass of sphere stays at rest.



Question:

Maximum kinetic energy attained by the sphere during the motion is :

(A) $\frac{1}{21}\pi R^4 \rho g$

(B) $\frac{2}{21}\pi R^4 \rho g$

(C) $\frac{3}{21}\pi R^4 \rho g$

(D) $\frac{4}{21}\pi R^4 \rho g$

SECTION 2 - Physics [MULTIPLE CORRECT TYPE]

Q.8

Three right angled prism of refractive index μ_1, μ_2 and μ_3 are joined together so that the faces of the middle prism are in contact each with one of the outside prisms. Choose the wrong option(s).

(A) If i is the incident angle and e is the emergent angle of the ray then for no net deviation e must be equal to $90 + i$.

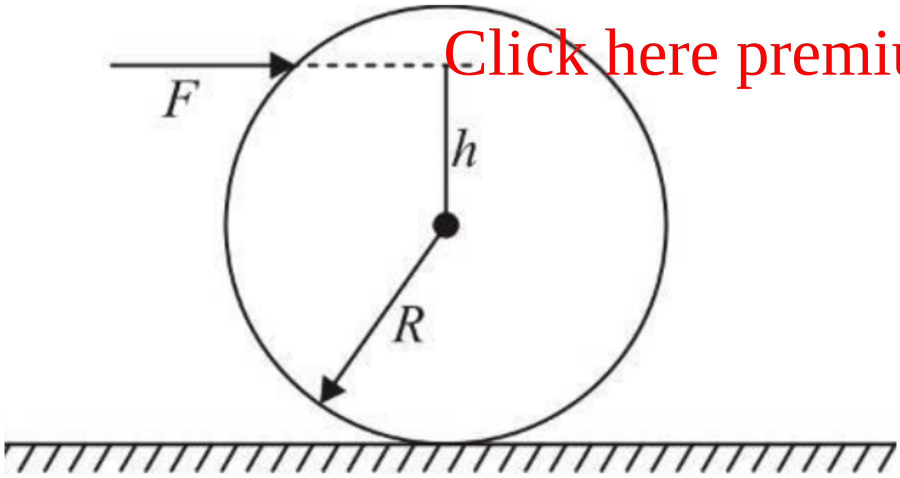
(B) If no net deviation of light is taking place through the combination then relation between μ_1, μ_2 and μ_3 must be ${\mu_1}^2 + {\mu_3}^2 - {\mu_2}^2 = 1$.

(C) If $\mu_2 > \mu_1$, no light will emerge out of the system.

(D) Net deviation of light through this combination can not be zero.

Q.9

A solid sphere of mass M and radius R is placed on a rough horizontal surface. A horizontal force F is applied to the sphere at a height $h (0 \leq h \leq R)$ from the centre as shown in figure. If the sphere rolls without slipping, then choose correct options.



(A) The maximum possible acceleration of the sphere is F/M

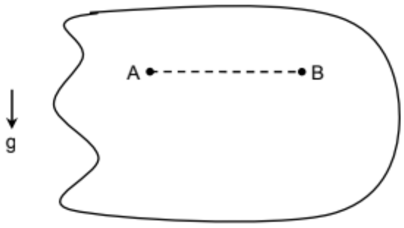
(B) The maximum possible acceleration of the sphere is $10F/7M$

(C) The magnitude of maximum possible friction force acting on the sphere is $3F/7$

(D) The magnitude of maximum possible friction force action on the sphere is equal to F.

Q.10

Consider a planar body of irregular size. It is kept in equilibrium in vertical plane by using 2 pins A& B lying at same vertical level as shown in figure.



Separation between $A\&B$ is d. Pins offer a small friction to the body. Normal contact offered to the body by pin A is N_A and by pin B is N_B. It is observed that when pin A is removed body rotates 150° anti-clockwise w.r.t. B and attains equilibrium. In the same manner if pin B is removed body rotates 120° clockwise w.r.t. A and attains equilibrium. Choose the correct option(s) :

(A) Value of $\frac{N_B}{N_A}$ is 3

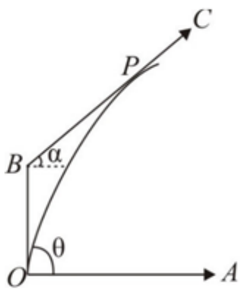
(B) Value of $\frac{N_B}{N_A}$ is 2

(C) Separation between centre of mass of body and pin A is $\frac{\sqrt{3}d}{2}$

(D) Separation between centre of mass of body and pin A is $\frac{d}{2}$

Q.11

A particle is projected from a point O in a horizontal surface OA with speed u and angle of projection θ. It just grazes the plane BC which make an angle α with the horizontal as shown in figure. The time taken by the projectile to reach P from the instant of projection is:



(A) $\frac{2u\sin\theta}{g\cos\alpha}$

(B) $\frac{u\sin\theta}{g\cos\alpha}$

(C) $\frac{u\sin(\theta-\alpha)}{g\cos\alpha}$

(D) $\frac{u\cos(\alpha-\theta)}{g\cos\alpha}$

Q.12

A point charge $\mathrm{q}=6\mu\mathrm{C}$ is moving in a straight line with a velocity $\vec{\mathrm{v}}=5\times10^{4}\vec{\mathrm{i}}(\mathrm{m}/\mathrm{s})$. When the charge is at the location $\mathrm{P}(3\ \mathrm{m}, 4\ \mathrm{m}, 0)$ choose the correct statements about the electric & magnetic fields produced by the charge at the origin.

(A) Magnitude of magnetic field is $9.6\times10^{-10}\ \mathrm{T}$

(B) Magnetic field is in $-\mathrm{z}$ direction

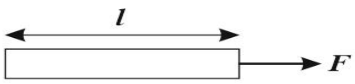
(C) Electric field is varying with time

(D) Magnetic field is decreasing (with time) in magnitude

SECTION 3 - Physics [NUMERICAL VALUE TYPE]

Q.13

A uniform rod of length l is acted upon by a force F in a gravity-free region, as shown in the figure. If the area of cross-section of the rod is A and it's Young's modulus is Y, then the elastic potential energy stored in the rod due to elongation is $U=\frac{F^2 l}{\mathrm{n}AY}$. Then find the value of n?



Q.14

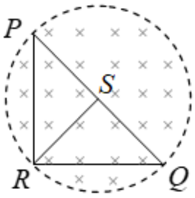
A radionuclide with disintegration constant λ is produced in a reactor at a constant rate $\alpha(=2\lambda)$ nuclei per second. At $t=0$, there are no nuclei present in the reactor. During each decay energy E_0 is released. 20% of this energy is utilized in increasing the temperature of water. The increase in temperature of m mass of water in time $t=T_{1/2}$ if specific heat of water is s is $772n$ K. Assume that there is no loss of energy through water surface. (Given $E_0=100$ times the energy required to raise the temperature of mass m of water from 0°C to 100°C) Find the value of n. (nearest integer)

Q.15

A 3.5 V mobile phone battery can produce 1 A of current for 1 hour. This can be charged using a square solar panel 25 cm on each side. Assuming an efficiency of 10% and an incident solar power of $1\mathrm{kWm}^{-2}$ what time (in minutes) is needed to charge the battery?

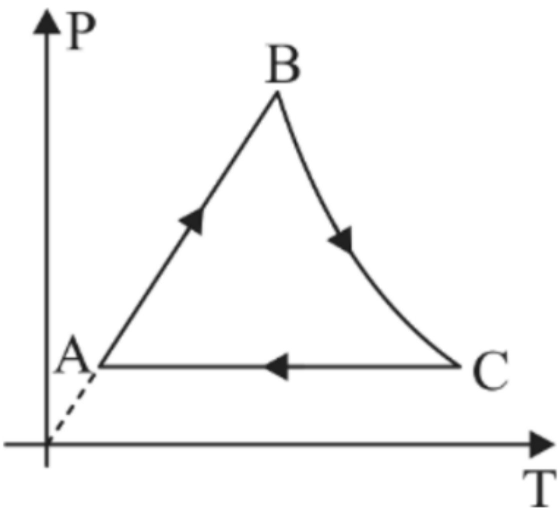
Q.16

PQR is a triangular frame made of a uniform metallic wire. RS is a median of the triangular frame joining the vertex R to the mid point S of the wire PQ. (Wire RS is also made of the same metallic wire). It is given that $PR=6$ cm, $RQ=8$ cm and $PQ=10$ cm. The cross sectional area of the wire is 1 mm^2 and its resistivity is $24\times10^{-9}\Omega m$. The triangle PQR lies in a cylindrical region of magnetic field such that the intersection of the surface of the cylinder with the plane containing the frame forms the circum circle of triangle PQR. The magnetic field in the region decreases at the rate of $0.526\ \mathrm{T}/\mathrm{s}$. The magnitude of the induced current in mA in the median RS of the frame is $5n$. Find n.



(A)
Q.17

A monoatomic gas initially in state A $(P_0,\ V_0)$ is taken through a cyclic process ABCA as shown in the figure. The pressure of point B is twice of that of A. The curve BC is a rectangular hyperbola. If the net work done by the gas is $xP_0\ V_0$, then find the value of x.



(A)
Q.18

A nonconducting disc of radius R and uniform positive surface charge density σ is placed on the ground with its axis vertical. A particle of specific charge $\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma}$ is dropped along the axis of the disc from a height h. The value of h if the particle just reaches the disc is $\left(2\sqrt{\frac{R}{3}}\right)^m$. Determine the value of m.

(A)

SECTION 1 - Chemistry [SINGLE CORRECT TYPE]

Q.19

An aq. solution containing excess of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. Loss in mass of the solution is $[\mathrm{Cu} = 64, \mathrm{O} = 16]$

(A) 0.294gm

(B) 0.373gm

(C) 1.073gm

(D) 1.443gm

Q.20

Which of the following combination will liberate hydrogen gas?

(A) Pb metal and dil. HNO_3

(B) Cu metal and dil. H_2SO_4

(C) Al metal and aq. NaOH

(D) Zn metal and conc. H_2SO_4

Q.21

For a complex (d^6 - configuration) having $\Delta_0 = 25000\ cm^{-1}$ and $P = 15000\ cm^{-1}$, the crystal field stabilisation energy is:

(A) $30,000\ cm^{-1}$

(B) $-60,000\ cm^{-1}$

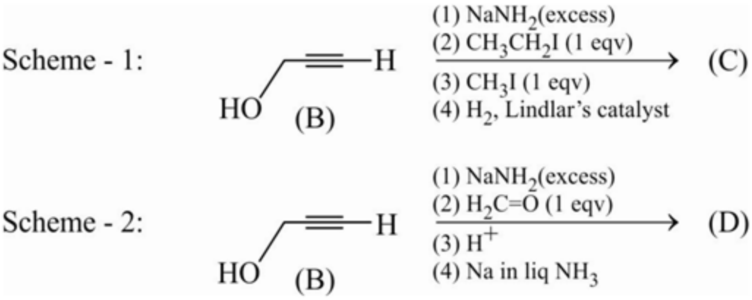
(C) $-30,000\ cm^{-1}$

(D) $-60,000\ cm^{-1}$

Q.22

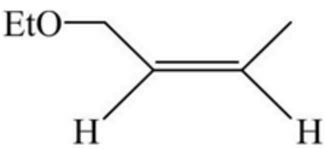
Paragraph:

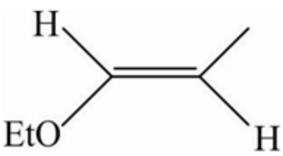
Schemes 1 and 2 describe sequential transformation of an alkyne (B) to (C) & (D) respectively. In each step of the transformation consider only the major product formed.

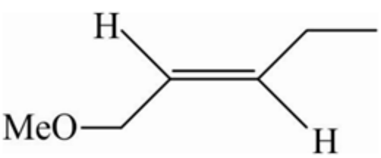


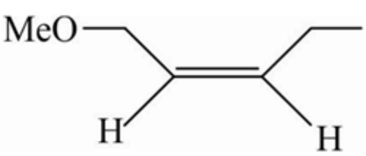
Question:

The product (C) would be :

(A) 

(B) 

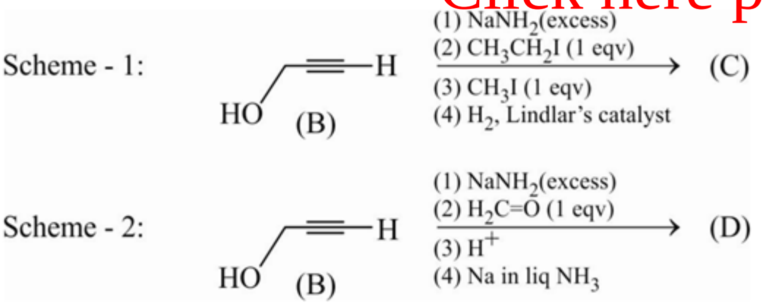
(C) 

(D) 

Q.23

Paragraph:

Schemes 1 and 2 describe sequential transformation of an alkyne (B) to (C) \& (D) respectively. In each step of the transformation consider only the major product formed.

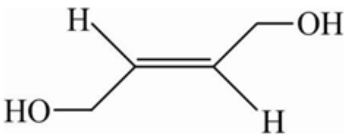


Question:

The incorrect statement with respect to product (D) is :

(A) It is identical to product (C) in scheme-1.

(B) The product (D) is



(C) The product (D) on oxidative ozonolysis gives 2 moles of HO_2CCH_2OH

(D) The product (D) on reaction with D_2 in presence of Ni gives a racemic mixture

Q.24

Paragraph:

$$\begin{aligned}&FeCr_2O_4 + NaOH + \text{ air } \rightarrow (A) + Fe_2O_3\\&(A) + (B) \rightarrow Na_2Cr_2O_7\\&Na_2Cr_2O_7 + X \xrightarrow{\Delta} Cr_2O_3\\&Cr_2O_3 + Y \xrightarrow{\Delta} Cr\end{aligned}$$

Compounds (A) and (B) are:

(A) Na_2CrO_4, H_2SO_4

(B) $Na_2Cr_2O_7, HCl$

(C) Na_2CrO_5, H_2SO_4

(D) $Na_4[Fe(OH)_6], H_2SO_4$

Q.25

Paragraph:

$$\begin{aligned}&FeCr_2O_4 + NaOH + \text{ air } \rightarrow (A) + Fe_2O_3\\&(A) + (B) \rightarrow Na_2Cr_2O_7\\&Na_2Cr_2O_7 + X \xrightarrow{\Delta} Cr_2O_3\\&Cr_2O_3 + Y \xrightarrow{\Delta} Cr\end{aligned}$$

(X) and (Y) are:

(A) C and Al

(B) Al and C

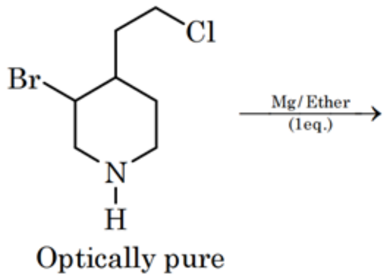
(C) C in both

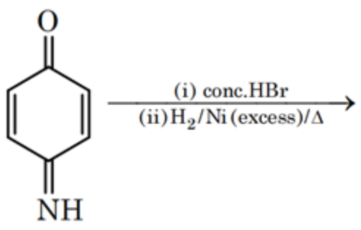
(D) Al in both

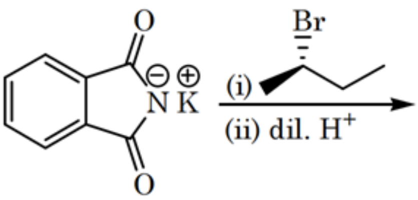
SECTION 2 - Chemistry [MULTIPLE CORRECT TYPE]

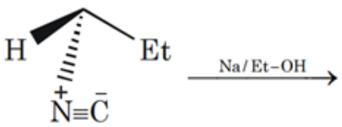
Q.26

Identify which reaction(s) produce(s) single optically active Amine molecule as product

(i) 

(ii) 

(iii) 

(iv) 

(A) ii, iii, iv

(B) iii only

(C) iii , iv

(D) none of these

Q.27

Assuming all non-hydrogenic atoms undergo hybridization \& following are various categories of orbital overlapping.

$\mathrm{sp}^3 - \mathrm{sp}^3$ axial ovelap $(\sigma) \longrightarrow$ Type $-$ A
$\mathrm{sp}^2 - \mathrm{sp}^2$ axial overlap $(\sigma) \longrightarrow$ Type $-$ B
$\mathrm{sp}^3 - \mathrm{sp}^2$ axial overlap $(\sigma) \longrightarrow$ Type $-$ C
$\mathrm{p} - \mathrm{p}$ lateral ovelap $(\pi) \longrightarrow$ Type $-$ D
$\mathrm{p} - \mathrm{d}$ lateral overlap $(\pi) \longrightarrow$ Type $-$ E

Identify the species which contain only 2 of these types (from A to E) of overlapping of orbitals.

(A) H_2CO_3

(B) $B_3 N_3H_3 F_3$

(C) SNF_3

(D) $H_2 S_4O_6$

Q.28

Irreversible process is :

$$\begin{aligned}&H_2O\,(\ell, 1\text{ atm}, -10^\circ C) \rightarrow H_2O\,(S, 1\text{ atm}, -10^\circ C)\\ &\text{Given at } 0^\circ C \quad \Delta H_{fusion} = 6\text{ kJ/mole}\\ &\text{at } -10^\circ C \quad \Delta H_{fusion} = 5.55\text{ kJ/ mole}\\ &C_p(\text{ ice }) = 30\text{ J/K-mole}\\ &C_p(\text{ water }) = 75\text{ J/K-mole}\\ &\ln\frac{273}{263} = 0.04\end{aligned}$$

Pick out the incorrect statements :

(A) ΔS_{total} for process $= -20.4$ J/K-mole

(B) ΔS_{surr} for process $= 21.1$ J/K-mole

(C) ΔS_{total} for process $= 0.92$ J/K-mole

(D) ΔS_{total} for process $= 22$ J/K-mole

Q.29

A lead storage cell is discharged which causes the H_2SO_4 electrolyte to change from a concentration of 40% by weight (density $= 1.260$ gm/ml) to 28%, by weight. The original volume of electrolyte was one litre. Identify the correct statement(s) :-

(A) The overall cell reaction is: $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(\ell)$.

(B) A total of 2.0 moles of H_2SO_4 have reacted.

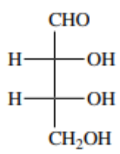
(C) The total charge released from anode of the cell is 1.93×10^5 coulomb.

(D) The mass of electrolytic solution has decreased.

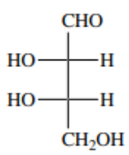
Q.30

Which of the following compounds on reduction give meso-1, 2, 3, 4-butanetetrol

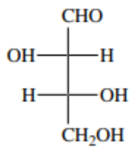
(A)



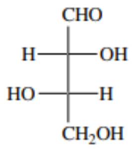
(B)



(C)



(D)



SECTION 3 - Chemistry [NUMERICAL VALUE TYPE]

Q.31

A hydrogen electrode placed in an aqueous solution of 0.1 M BCl (a salt of weak base and strong acid) and 10.0 M BCl has oxidation potential values 0.115 V and 0.180 V at 25°C. What is the pK_a value of B^+?

(A)

Q.32

One mole of a real gas obeys $P(Vm - b) = RT$; where ' b ' and ' R ' are constants with usual meanings. If occupied volume of the gas is ' 10 b ' then the compressibility factor of gas will be :

(A)

Q.33

How many of the following compounds are amphoteric in nature?

$PbO, PbO_2, SnO, SnO_2, Al_2O_3, ZnO, BeO, Ga_2O_3,\ B_2O_3$.

(A)

Q.34

140 day old radioactive matter has an activity of 6000 dps.70 days later it shows an activity of 2000 dps. Then ratio of its initial activity to activity when it was 140 day old is ________.

(A)

Q.35

A compound having molecular formula $C_5H_{10}O_5$ was reacted with excess of CH_3COCl to give compound having molecular weight 318 gm. Predict the number of hydroxy groups present into the unknown compound.

(A)

Q.36

In fusion reaction of chromite ore with sodium carbonate in free access of air, ' a ' number of elements undergo redox change \& ' b ' moles of O_2 are consumed per mole of chromite ore. Report $(a + b)$

(A)

SECTION 1 - Mathematics [SINGLE CORRECT TYPE]

Q.37

Let A, B and C be 3 square matrices of order 3×3. If $A^3 - 6A^2 + 7A + B = 0$ and $(\operatorname{adj}(\operatorname{adj} B)) = C$, where $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then

$\left(\frac{\text{sum of the elements of } C}{10}\right)$ is

(A) 2.4

(B) 4.8

(C) 3.2

(D) 6.4

Q.38

Let $\mathrm{f}''(\mathrm{x})<0 \quad \forall \mathrm{x} \in \mathrm{R}$ and $\mathrm{g}(\mathrm{x})=\mathrm{f}\left(2-\mathrm{x}^{2}\right)+\mathrm{f}\left(\mathrm{x}^{2}\right)$ then set of possible values of x for which g(is increasing is :

(A) $(-1,0) \cup(1, \infty)$

(B) $(-\infty,-1) \cup(1, \infty)$

(C) $(-\infty,-1) \cup(0,1)$

(D) $(-1,1)$

Q.39

If $\mathrm{x}^{8}-(\mathrm{k}-1) \mathrm{x}^{4}+5=0$, then least possible integral value of ' k ' so that equation has maximum number of real roots is :

(A) 4

(B) 5

(C) 6

(D) 7

Q.40

Paragraph:

Let a point P whose position vector is $\vec{r}=x \hat{i}+y \hat{j}+z \hat{k}$ is called lattice point if $x, y, z \in N$. If atleast two of x, y, z are equal then this lattice point is called isosceles lattice point. If x, y, z are equal then this lattice point is called equilateral lattice point.

Question:

Which of following is (are) correct?

(A) The number of lattice point on the plane $\vec{r} \cdot(\hat{i}+\hat{j}+\hat{k})=10$ are 45

(B) The number of lattice point on the plane $\vec{r} \cdot(\hat{i}+\hat{j}+\hat{k})=10$ are 36

(C) Area of triangle formed by isosceles lattice points lying on the plane $\vec{r} \cdot(\hat{i}+\hat{j}+\hat{k})=4$ is $\frac{\sqrt{3}}{2}$

(D) Option 2 and 3

Q.41

Paragraph:

Let a point P whose position vector is $\vec{r}=x \hat{i}+y \hat{j}+z \hat{k}$ is called lattice point if $x, y, z \in N$. If atleast two of x, y, z are equal then this lattice point is called isosceles lattice point. If x, y, z are equal then this lattice point is called equilateral lattice point.

Question:

If a lattice point is selected at random from lattice points which satisfy $\vec{r} \cdot(\hat{i}+\hat{j}+\hat{k}) \leq n$, then the probability that the selected lattice point is equilateral given that it is isosceles lattice point is :

(A) $\frac{1}{21}$ is $n=11$

(B) $\frac{1}{22}$ is $n=11$

(C) $\frac{1}{17}$ is $n=10$

(D) $\frac{1}{18}$ is $n=10$

Q.42

Paragraph:

Let us define a function $g(x)=\begin{cases}x^2 & ;0\leq x<1/2\\ (x-1)^2; & \frac{1}{2}\leq x\leq 1\end{cases}$ and another function $f(x)$ such that the Area enclosed by curves $g(x)$ and x-axis between the ordinates $x=\beta$ and $x=\beta+f(\beta)$ is half the area enclosed by $g(x)$ and x-axis. (where $0<\beta<\frac{1}{2}$ and $\frac{1}{2}<\beta+f(\beta)<1$).

Question:

The value of $\lim_{x\to\frac{1}{2}}\frac{(f(x)+x-1)^3}{(x-\frac{1}{2})}$, is :

(A) $\frac{3}{4}$

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 1

Q.43

Paragraph:

Let us define a function $g(x)=\begin{cases}x^2 & ;0\leq x<1/2\\ (x-1)^2; & \frac{1}{2}\leq x\leq 1\end{cases}$ and another function $f(x)$ such that the Area enclosed by curves $g(x)$ and x-axis between the ordinates $x=\beta$ and $x=\beta+f(\beta)$ is half the area enclosed by $g(x)$ and x-axis. (where $0<\beta<\frac{1}{2}$ and $\frac{1}{2}<\beta+f(\beta)<1$).

Question:

Let at $x=\alpha$, $f(x)$ has a local extremum, then the value of α^3, is :

(A) 1/16

(B) 1/8

(C) 1/4

(D) 1

SECTION 2 - Mathematics [MULTIPLE CORRECT TYPE]

Q.44

Let A and B be two square matrices of order n satisfying $A+BA^T=I$ and $B+AB^T=I$ and O is a null matrix and I is the identify matrix of order n, then identify the correct statement(s)

(A) $\mathrm{A}^4-2\,\mathrm{A}^2+\mathrm{A}=\mathrm{O}$

(B) $\mathrm{A}^4-2\,\mathrm{A}^2-\mathrm{A}=\mathrm{O}$

(C) $\mathrm{A}^2+\mathrm{B}=\mathrm{I}$

(D) $\mathrm{B}^4-2\,\mathrm{B}^2-\mathrm{B}=\mathrm{O}$

Q.45

Let $\bar{x}$ and $\bar{y}$ be unit vectors. If $\bar{r}$ is a vector such that $\bar{r} + (\bar{r} \times \bar{x}) = \bar{y}$, then

(A) maximum value of $\bar{r} \cdot \bar{x}$ is 1

(B) maximum value of $|\bar{r} \cdot \bar{x} \times \bar{y}|$ is $\frac{1}{2}$

(C) maximum value of $|\bar{r}|$ is 1

(D) minimum value of $|\bar{r}|$ is $\frac{1}{\sqrt{2}}$

Q.46

Let $f : (0, \infty) - \{1\} \to R$ be a differentiable function such that $f'(x) = 1 - \frac{f(x)}{x-1}$ for all $x \in$ domain of the function and $f(2) \neq \frac{1}{2}$, then:

(A) $\lim_{x \to 0^+} f'\left(\frac{1}{x}\right) = \frac{1}{2}$

(B) $\lim_{x \to 0^+} x f\left(\frac{1}{x}\right) = \frac{1}{2}$

(C) $\lim_{x \to 0^+} x^2 f'(x) = 1$

(D) $|f(x)| \leq 2$ for all $x \in (0, 2)$

Q.47

If $\left|\frac{z-\alpha}{z-\beta}\right| = k, k > 0$ where, $z = x + iy$ and $\alpha = \alpha_1 + i\alpha_2, \beta = \beta_1 + i\beta_2$ are fixed complex numbers. Then which of the following are true

(A) if k $\neq 1$ then locus is a circle whose centre is $\left(\frac{\mathrm{k}^2\beta - \alpha}{\mathrm{k}^2 - 1}\right)$

(B) if $k \neq 1$ then locus is a circle whose radius is $\left|\frac{k(\alpha - \beta)}{1 - k^2}\right|$

(C) if $k = 1$ then locus is perpendicular bisector of line joining $\alpha = \alpha_1 + i\alpha_2$ and $\beta = \beta_1 + i\beta_2$

(D) if $k \neq 1$ then locus is a circle whose centre is $\left(\frac{k^2\alpha - \beta}{k^2 - 1}\right)$

Q.48

If $f : R \to [a, 6]$ defined by $f(x) = \frac{x^2 - 2x + d}{x^2 + 3x + d}$ is an onto function, then

(A) a is equal to $\frac{2}{7}$

(B) d is equal to 4

(C) a is equal to 1

(D) $d - 7a$ is equal to -3

SECTION 3 - Mathematics [NUMERICAL VALUE TYPE]

Q.49

Let E_1 and E_2 be two Ellipse. The area of the Ellipse E_2 is one-third the area of the quadrilateral formed by the tangents at the ends of the latus rectum of the ellipse E_3 $\left(E_3 : 5x^2 + 9y^2 = 45\right)$. The eccentricities of E_1, E_2 and E_3 are equal. E_1 is inscribed in E_2 in such a way that both E_1 and E_2 touch each other at one end of their common major axis. If the length of the major axis of E_1 is equal to the length of the minor axis of E_2, then find the area of the ellipse E_2 outside the ellipse E_1.

[illegible]

Q.50

Let $f(x) = (x-1)^{100}(x-2)^{2(99)}(x-3)^{3(98)} \ldots\ldots (x-100)^{100}$ where $k = \frac{f'(101)}{f(101)}$. Find the value of $\frac{k}{50} - 97$

[illegible]

Q.51

The line $y = x$ touches a circle at a point P in the first quadrant such that the distance of P from the origin is $5\sqrt{2}$. If the length of the portion intercepted by this circle on the y-axis is 5 and the equation of the circle, is $x^2 + y^2 + Ax + By + C = 0$. Then find the value of $\frac{|A+B+C|}{6}$.

Q.52

Suppose a, b, c are such that the curve $y = ax^2 + bx + c$ is tangent to $y = 3x - 3$ at $(1, 0)$ and is also tangent to $y = x + 1$ at $(3, 4)$. Then the value of $(2a - b - 4c)$ equals ________.

Q.53

A bag contains 3 red, 2 white and 2 black balls. Two balls are drawn at random and none of them is found to be a white ball. The probability that both balls are red is $\frac{a}{b}$ (where a, b are coprime) then $b - a$ is equal to

Q.54

Let T_n be the n^{th} term of a series for $n \geq 1$ and $T_1 = 1$, $T_{n+1} = 2T_n + 4^n, n \geq 1$. If $S_n =$ Sum of first n terms of the series and $S_n = \frac{a.4^n + b \cdot 2^n + 1}{3}$ where $a, b \in I$, then the value of $\left|\frac{b}{a}\right|$ is